Qian Wang

NY | 646-670-2955 | wangqian01222026@gmail.com | LinkedIn | GitHub | Google Scholar | Portfolio

No employment sponsorship required

RESEARCH PROFILE

Interdisciplinary scientist working at the interface of structural biology, protein biochemistry, and artificial intelligence. Brings more than nine years of experience in hypothesis-driven experimental research, with expertise in biomolecular NMR, protein–metal interactions, enzyme mechanisms, and scientific machine learning. Develops reproducible computational methods for protein function, structure, and biophysical-property prediction and translates fundamental biological questions into rigorous experimental and data-driven research programs.

EDUCATION

University of Pennsylvania *M.S. in Computer and Information Technology* **2025**

The City University of New York *Ph.D. in Biochemistry* **2017**

RESEARCH EXPERTISE

Structural Biology & Biochemistry: biomolecular NMR spectroscopy, protein expression and purification, protein–protein and protein–metal interactions, enzyme mechanisms, mutagenesis, biophysical assay development

AI for Biology & Drug Discovery: protein function prediction, protein structure analysis, sequence and structure embeddings, biological feature engineering, biophysical-property prediction

Scientific Machine Learning: Python, PyTorch, PyTorch Geometric, BioPython, graph neural networks, attention models, protein language models, ESM-2, supervised learning, custom loss functions, held-out evaluation, reproducible ML pipelines

Research Leadership: experimental design, grant-supported research, workshop and curriculum development, laboratory program administration, trainee mentorship, interdisciplinary collaboration

SELECTED AI & COMPUTATIONAL RESEARCH PROJECTS

Scientific Manuscript Writing Agent **2026**

Claude API, agentic workflows, scientific automation, retrieval-grounded generation

- Designed an 11-stage AI-assisted workflow that transforms experimental notes and figures into structured manuscript drafts while preserving scientist review at key decisions.
- Engineered stage-gated orchestration with persistent checkpoints and resumable state, limiting error propagation and enabling traceable recovery across long-running workflows.
- Built retrieval-grounded drafting, figure-caption generation, and scientific tone-transfer modules using expert-curated instructions to improve consistency and traceability.

Protein Function Prediction using ESM-2 + GCN **2025–2026**

Protein language models, graph convolutional networks, Kaggle CAFA6

- Developed an end-to-end protein-function prediction pipeline combining ESM-2 sequence embeddings with graph neural networks for the CAFA6 community benchmark.
- Implemented reproducible data preparation, model training, and held-out evaluation workflows for multilabel biological prediction and challenging protein-level generalization.
- Translated protein annotations and biological relationships into machine-readable features and model-design decisions, connecting domain science with ML engineering.

NMR-HSQC-GNN: Protein Chemical Shift Predictor **2024–Present**

PyTorch Geometric, structural biology, graph neural networks, ML evaluation

- Developed a GATv2 graph neural network using 4,286 paired BMRB/PDB entries to predict backbone ^{1}H and ^{15}N secondary chemical shifts directly from protein 3D structures.
- Improved Pearson correlation from 0.43 to 0.74 through spatial and hydrogen-bond graph edges; achieved test MAE of 0.302 ppm for ^{1}H and 1.749 ppm for ^{15}N across 620 held-out proteins.
- Designed a referencing-invariant pairwise loss inspired by crystallographic Patterson analysis to reduce label noise during model training and improve robustness to systematic dataset artifacts.

RESEARCH EXPERIENCE

Scientific Program Administrator & Research Mentor, NIH G-RISE Program **2026–Present**

The City College of New York, New York, NY

Graduate Research Training Initiative for Student Enhancement (G-RISE)

- Administer NIH-supported research-training funds and coordinate G-RISE activities across multiple biomedical research laboratories, aligning program resources with trainee-development and scientific-training objectives.
- Develop the Research Identity Bootcamp and related workshop curricula in consultation with faculty mentors and laboratory leadership, with emphasis on research design, scientific communication, and professional development.
- Provide direct, individualized mentorship to Ph.D. trainees, strengthening research practice, technical decision-making, and readiness for independent biomedical research careers.

Research Associate **2017–2026**

The City University of New York, Staten Island, NY

- Led hypothesis-driven studies of difficult-to-express proteins, transient intermediates, and protein–protein interactions using cloning, mutagenesis, purification optimization, and NMR characterization.

- Increased soluble protein yield 10-fold through systematic workflow optimization, enabling downstream structural and functional analysis of a previously unreliable target.
- Supported more than $1M in research funding by rapidly generating robust preliminary data and translating complex scientific questions into literature-backed proposal concepts.
- Integrated literature analysis, experimental data, and biophysical interpretation to guide research priorities and communicate findings to scientific collaborators and funding stakeholders.
- Mentored Ph.D. and undergraduate researchers in experimental design, protein science, spectroscopy, and scientific communication, supporting on-time Ph.D. completion and two campus-level trainee awards.

PEER-REVIEWED PUBLICATIONS

1. Kothalawala, M., Shirazi, S., **Wang, Q.**, Collado, A., Bongiorno, A., & Gupta, R. (2025). Molecular basis of calcium-induced acidic shift in antimicrobial zinc sequestration by S100A12. *Journal of Physical Chemistry B, 129*, 9929–9938. doi:10.1021/acs.jpcb.5c04464
2. **Wang, Q.**, Aleshintsev, A., Rai, K., Jin, E., & Gupta, R. (2024). Proton transfer via arginine with suppressed pK_a mediates catalysis by gentisate and salicylate dioxygenase. *Journal of Physical Chemistry B, 128*, 6797–6805. doi:10.1021/acs.jpcb.4c03164
3. **Wang, Q.**, DiForte, C., Aleshintsev, A., Elci, G., Bhattacharya, S., Bongiorno, A., & Gupta, R. (2024). Calcium-mediated static and dynamic allostery in S100A12: Implications for target recognition by S100 proteins. *Protein Science, 33*, e4955. doi:10.1002/pro.4955
4. **Wang, Q.**, Li, H., Bujupi, U., Gröning, J., Stolz, A., Bongiorno, A., & Gupta, R. (2024). Oxygen activation in aromatic ring-cleaving salicylate dioxygenase: Detection of reaction intermediates with a nitro-substituted substrate analog. *ChemBioChem, 25*, e202400023. doi:10.1002/cbic.202400023
5. **Wang, Q.**, Kuci, D., Bhattacharya, S., Hadden-Perilla, J. A., & Gupta, R. (2022). Dynamic regulation of Zn(II) sequestration by calgranulin C. *Protein Science, 31*, e4403. doi:10.1002/pro.4403
6. **Wang, Q.**, Aleshintsev, A., Jose, A. N., Aramini, J. M., & Gupta, R. (2020). Calcium regulates S100A12 zinc sequestration by limiting structural variations. *ChemBioChem, 21*, 1372–1382. doi:10.1002/cbic.201900623
7. **Wang, Q.**, Aleshintsev, A., Bolton, D., Zhuang, J., Brenowitz, M., & Gupta, R. (2019). Ca(II) and Zn(II) cooperate to modulate the structure and self-assembly of S100A12. *Biochemistry, 58*, 2269–2281. doi:10.1021/acs.biochem.9b00123
8. **Wang, Q.**, Rizk, S., Bernard, C., Lai, M. P., Kam, D., Storch, J., & Stark, R. E. (2017). Protocols and pitfalls in obtaining fatty acid-binding proteins for biophysical studies of ligand–protein and protein–protein interactions. *Biochemistry and Biophysics Reports, 10*, 318–324. doi:10.1016/j.bbrep.2017.05.001